Visual and Auditory Systems

MED 808: Neuroscience, Neuroanatomy, and Neurology (NNN), Fall 2026

Reading:

Linda S. Costanzo: Physiology, 8th Ed., (Refer to Course Syllabus).

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Learning Objectives

After studying this lecture, the student should be able to:

1. Explain the mechanisms of phototransduction in retinal rod and cone photoreceptors and predict how light- and dark-induced changes in photopigments, cyclic GMP, ion-channel activity, and neurotransmitter release influence bipolar and ganglion cell signaling.
2. Describe the structure of the outer, middle, and inner ear, and relate their anatomical organization to hearing and balance.
3. Explain the organization and function of the auditory and vestibular pathways from the inner ear to the cerebral cortex.

Lecture Outline

I. Photoreceptors

II. Photoreception in the Rod Cell

III. Vision Field Defects

IV. Overview of Hearing

V. External and Middle Ear: Sound Conduction

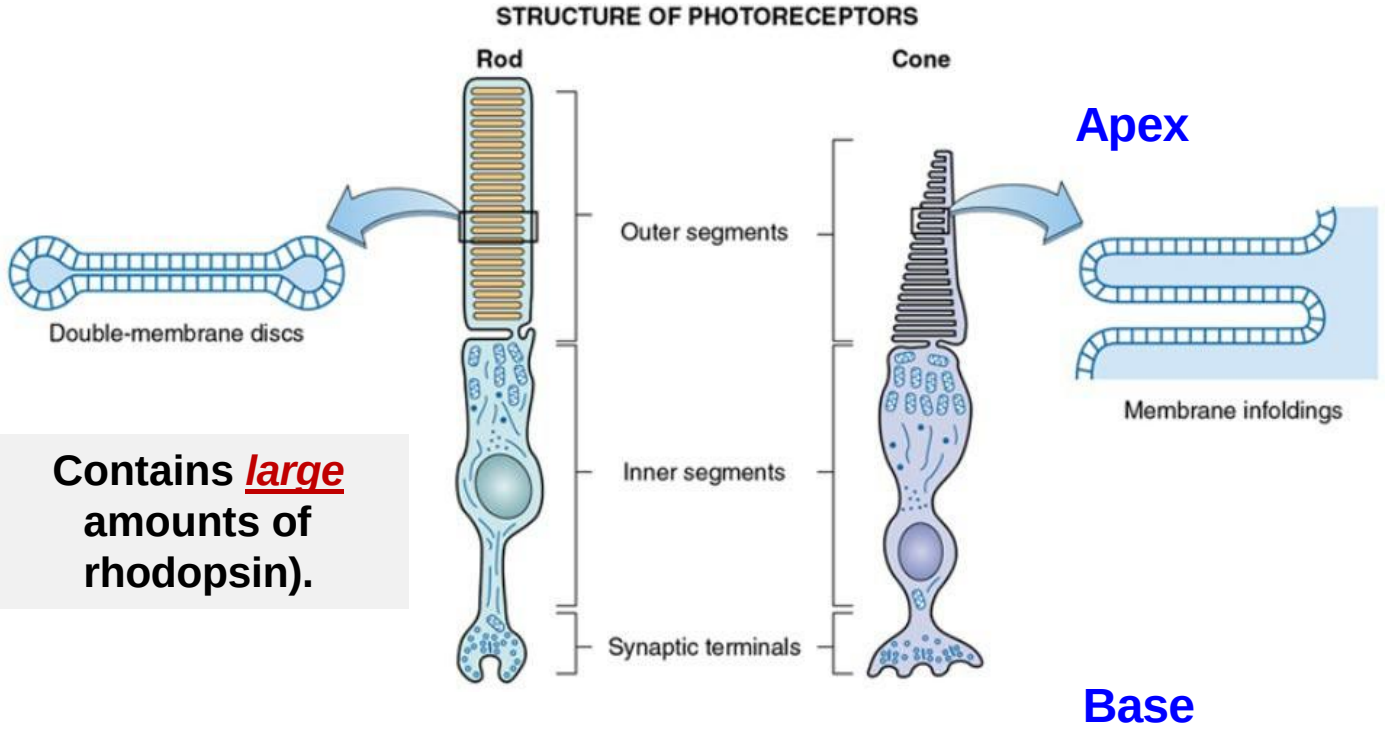
VI. Physiological Mechanism of Auditory Signal Transduction

VII. Auditory Neural Pathway

Photoreceptors

Photoreceptors

- **Two types of photoreceptors:**
  - Rods
  - Cones
- Modified neurons
  - **Outer segment:** detects light
  - **Inner segment:** cell body
  - **Synaptic terminal:** connects to interneurons
- **Photoreceptors** located in retina.



Contains *large* amounts of rhodopsin).

NIGHT vision
HIGH sensitivity to light
LOW visual acuity

COLOR vision
LOW sensitivity to light
HIGH visual acuity

FIG. 3.15 Structure of photoreceptors. The enlargements show a magnified view of the outer segments. Outer segment contains rhodopsin (a light-sensitive pigment). The greater the amount of photopigment, the greater the sensitivity to light. The inner segments contain mitochondria and other organelles.

TABLE 3.4 Properties of Rods and Cones

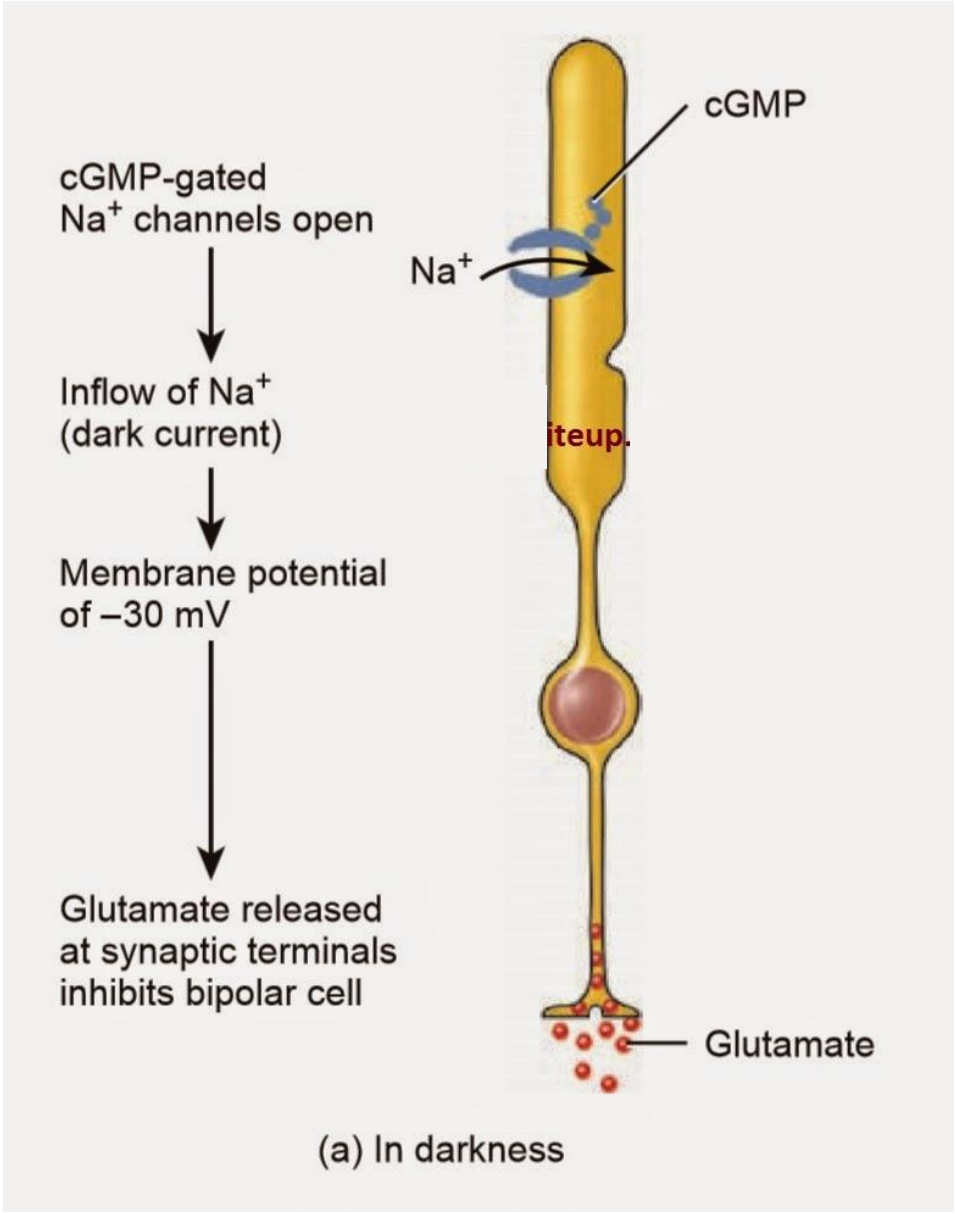
Photoreceptor	Sensitivity to Light	Acuity	Dark Adaptation	Color Vision
Rods	Low threshold Sensitive to low- intensity light Night vision	Low acuity Not present on fovea	Adapt late	No
Cones	High threshold Sensitive to high- intensity light Day vision	High acuity Present on fovea	Adapt early	Yes

Photoreception in the Rod Cell

Photoreception

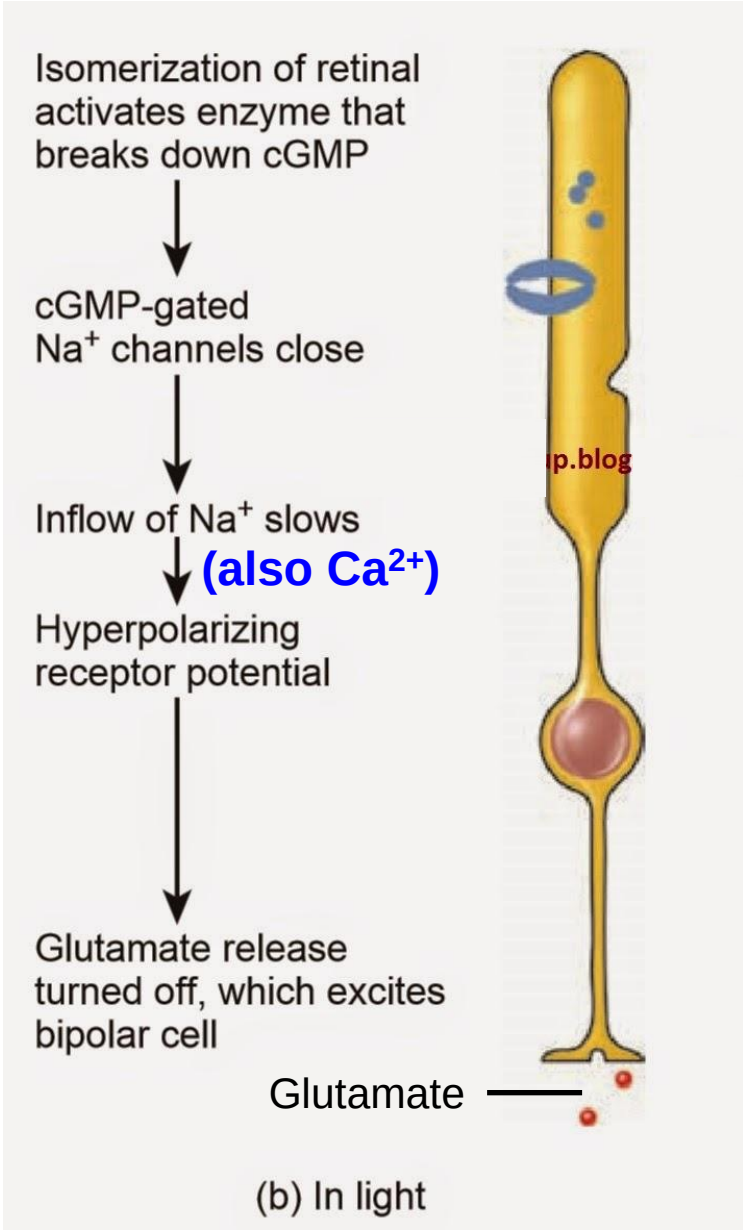
- Both **rhodopsin** and **opsin** are essential proteins involved in the phototransduction process, but they serve different roles in vision.
- **Rhodopsin structure:**
  - **Visual pigment** in **rod cells** of the retina
  - Composed of **11-*cis*-retinal** (a light-sensitive chromophore derived from Vitamin A) bound to **opsin** (a protein)
  - **In the dark, rhodopsin** is stable and **has a purple color** due to the specific configuration of **11-*cis*-retinal**

1. **In darkness,** the **cyclic GMP levels** of photoreceptors are **high**.
2. This level of cGMP **opens the cGMP ligand-gated Na^+ ions channels**.
3. **Inflow of Na^+ ions** through the channel in the photoreceptor takes place called as **"dark current"** which **partially depolarizes the photoreceptor**
   .
4. As a result, in darkness the **membrane potential** of a **photoreceptor is -30 mV** which is much closer to -70 mV.
5. This partial depolarization during darkness **triggers the release of neurotransmitters** at the synaptic terminals.
6. The neurotransmitter is the amino acid **glutamate** or **glutamic acid**.
7. This **glutamate inhibits the bipolar cells** that synapse with rods.



REACTION OF PHOTORECEPTOR IN DARKNESS

1. When **light** strikes on the retina the **cis-retinal** undergoes **isomerization** to **trans-retinal**, enzyme transduction get activated which activates **phosphodiesterase** which breaks down the **cyclic GMP** and leads to **decrease in cyclic GMP levels** which in turn is responsible for **closing** of **cGMP-gated Na^+ and Ca^{2+} channels**.
2. So, **inflows of Na^+ and Ca^{2+} ions** in the photoreceptor get **decreases**.
3. The **membrane potential** becomes more negative **-70 mV** (**hyperpolarizing** receptor potential).
4. **Decreases the release of neurotransmitter glutamate**.
5. Thus, **light excites bipolar cells that synapse with rods** by **turning off the release of inhibition of neurotransmitter**.
6. The **excited bipolar cells** subsequently **stimulate the ganglion cells** to form **action potentials in their axons**.



REACTION OF PHOTORECEPTOR IN LIGHT

ARRISTIN BINDS HERE
After rhodopsin kinase phosphorylates and activates rhodopsin.
Prevents further transducin activation.

- **Phototransduction in retinal rod cells in response to light is triggered by capture of photon by opsin (rhodopsin absorbs a photon)** → changing **opsin** conformation.
- **Metarhodopsin II** (represents **activated rhodopsin** after phosphorylation by rhodopsin kinase).
- **Activated rhodopsin (= metarhodopsin II**) binds to the G protein **transducin**.
- **Transducin dissociates and activates a cGMP phosphodiesterase** → hydrolyses cGMP to **5'-GMP** → ↑ **breakdown of cGMP** → ↓ cGMP levels → **closing cGMP-gated Na^+/Ca^{2+} channels** (↓ Na^+ and Ca^{2+} influx) → reversing the depolarization → enhancing **light-induced hyperpolarization (hyperpolarizing photoreceptor membrane potential** to **~ –70 mV).**

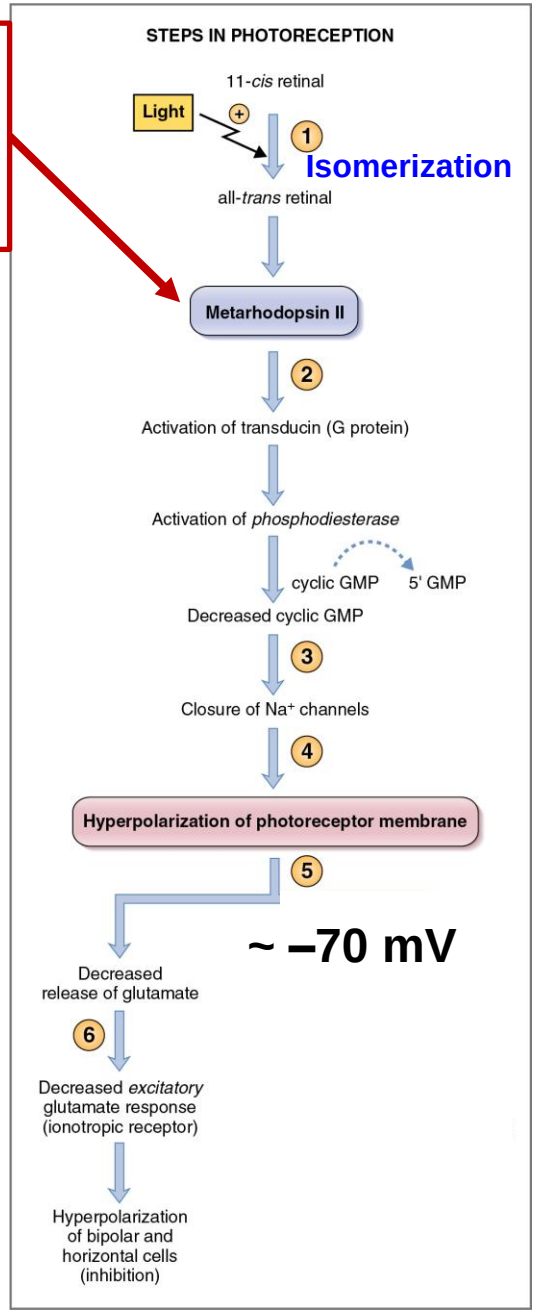


FIG. 3.16. Steps in photoreception. When **light** impinges on the retina, the photoreceptors are hyperpolarized. In turn, the photoreceptors **decrease their release of glutamate**, leading to either **hyperpolarization** or **depolarization** of **bipolar or horizontal cells** (i.e., decreased excitation). Circled numbers correlate with steps described in the text. ***Cyclic GMP,*** Cyclic guanosine monophosphate; ***GMP,*** guanosine monophosphate. **Note: Metarhodopsin II** is the light-excited form of rhodopsin that triggers G protein function.

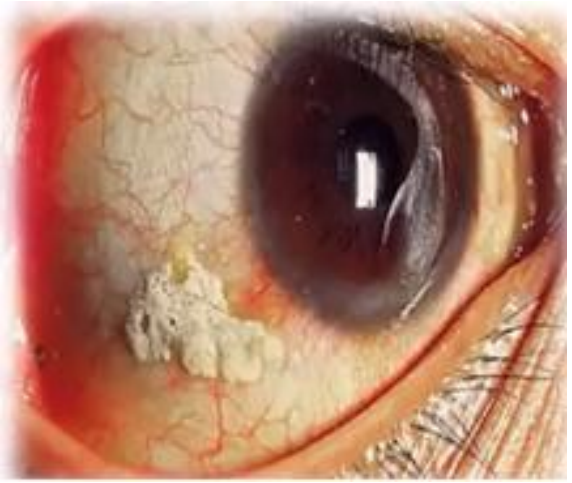
Vision in the Presence of Darkness and Light

- **Neurotransmitter glutamate is continuously released**
  - Because membrane is partially depolarized in the dark
  - **Glutamate *inhibits*** the **optic nerve bipolar cells** with which the rod cells synapse
- **Light hyperpolarizes the rod cell membrane → stops release of glutamate**
  - Relieves ***inhibition*** of optic nerve bipolar cell
  - Initiates signal into brain
- The **rod cell** return to its normal state quickly when **activated rhodopsin is inactivated**, and a protein called **arrestin subsequently binds it**.

Clinical Correlation: Vitamin A Deficiency and Night Blindness

- **Definition:**
  - **Night blindness (nyctalopia)** occurs with **severe vitamin A deficiency**
  - Due to **inadequate formation of retinal and rhodopsin** in photoreceptor cells
- **Pathophysiology:**
- **Vitamin A → retinal → rhodopsin** (light-sensitive pigment in rods)
  - **Deficiency** → ↓ rhodopsin synthesis → impaired **rod function**
  - **In low-light conditions, rods cannot** generate sufficient visual signals
- **Clinical Manifestation:**
  - **Poor vision in dim light or at night**
  - Vision may be normal in bright daylight (cones unaffected initially)

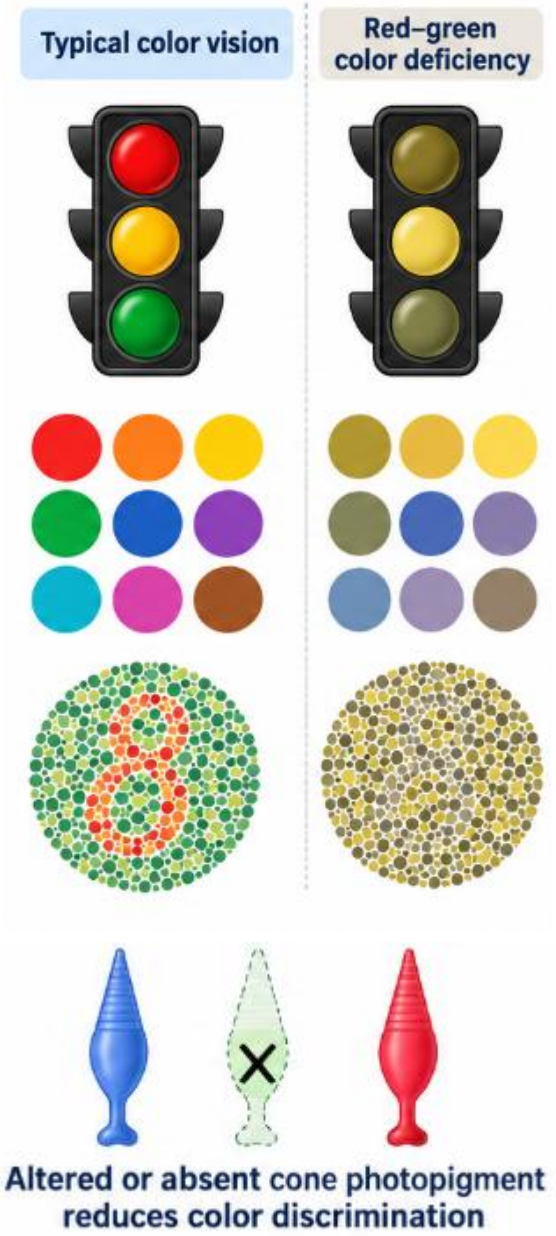
Clinical Correlation: Vitamin A Deficiency and Night Blindness, Cont.



- **Blind spots:**
  - Spots or patches in the eye
- **Time Course:**
  - Requires **months of vitamin A deficiency** to deplete **liver stores**
  - Liver normally contains **large vitamin A reserves** that supply the retina
- **Treatment and Reversal:**
  - **Rapid improvement** possible with **intravenous vitamin A**
  - **Rhodopsin synthesis restored** within **minutes to an hour** after supplementation

Clinical Correlation: Color Blindness

- **Color vision** depends on **cone photoreceptors** containing **photopsin** with different wavelength sensitivities.
- The common **red-green color-vision deficiencies** are usually **X-linked abnormalities of cone opsin genes** → impairment of color discrimination.
- **Abnormal or absent cone opsin** causes **inherited color blindness** = **inherited color-vision deficiency**.
- **Usually congenital**, bilateral, and nonprogressive.
- **Visual acuity** is usually **normal**.
- **NOTE: Rods** mediate **dim-light vision** and **DO NOT** distinguish color.



Visual Perception

Visual Perception

- **Definition:**
  - **Visual perception** is the process by which the **brain interprets and organizes visual information** received from the eyes, transforming light patterns into meaningful representations — such as objects, faces, motion, **depth**, and color.
  - **In short: Vision ≠ perception** — vision captures the image; **perception gives it meaning**.

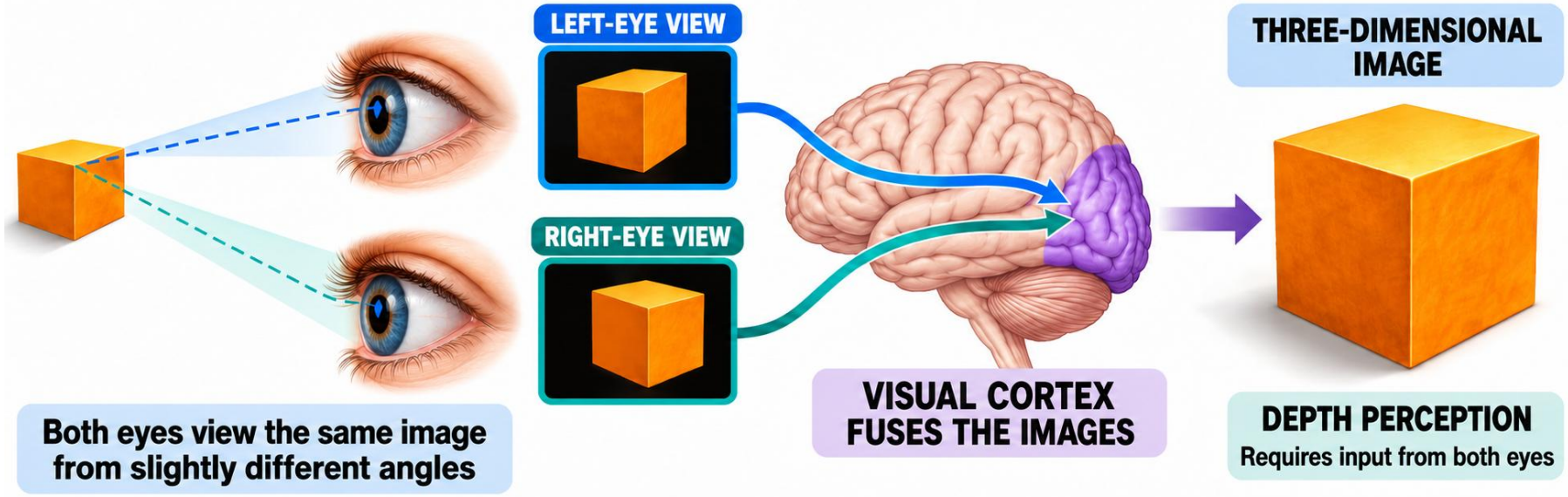
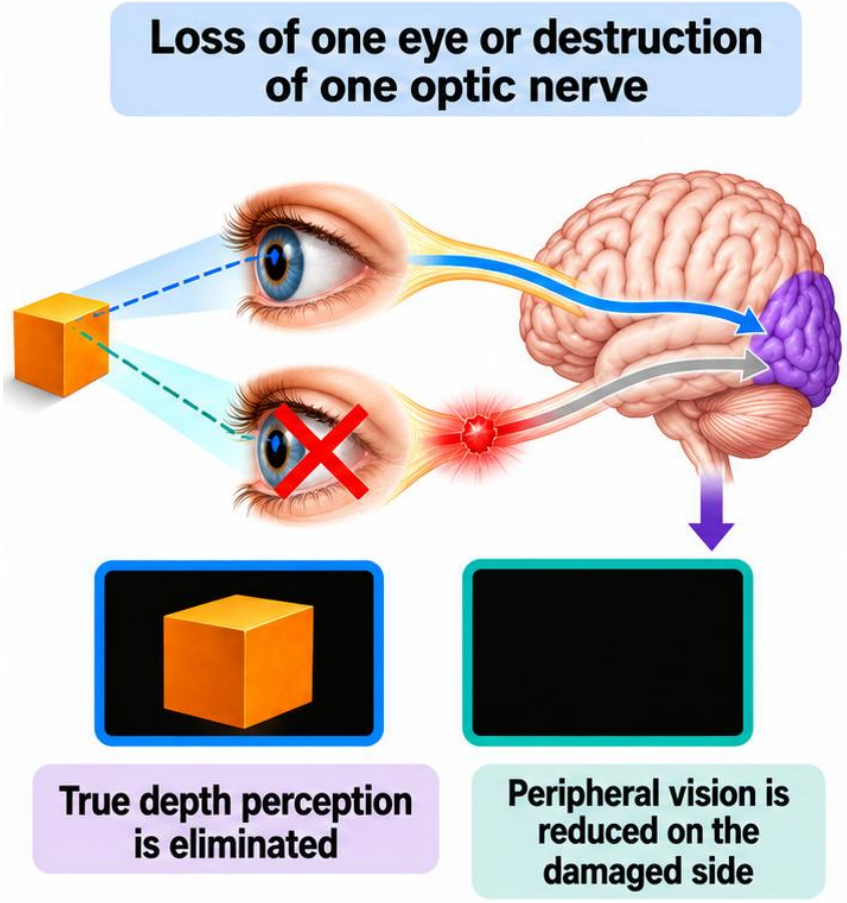
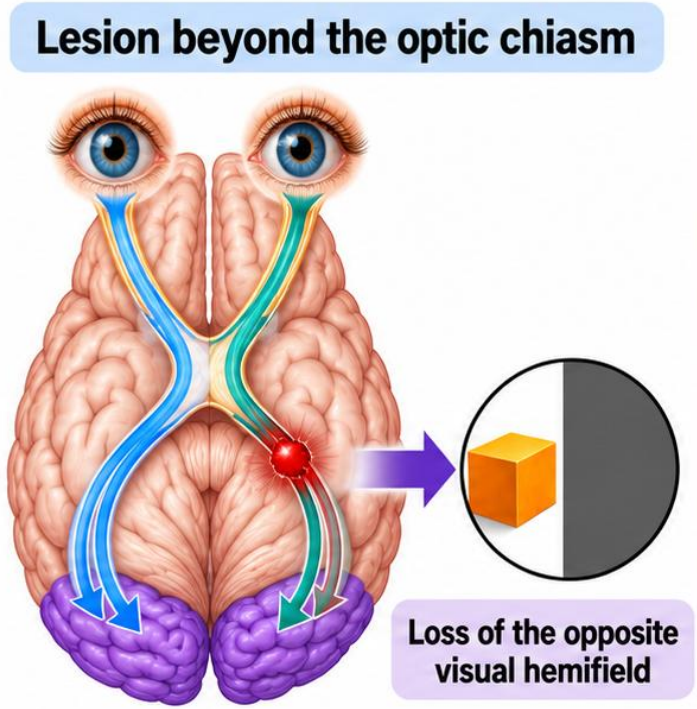


Image Illustrating Depth Perception.

- Both eyes view same image from slightly different angles.
- **Visual cortex** fuses these slightly different images, resulting in a three-dimensional image, which leads to **depth perception**.
- Requires input from both eyes.





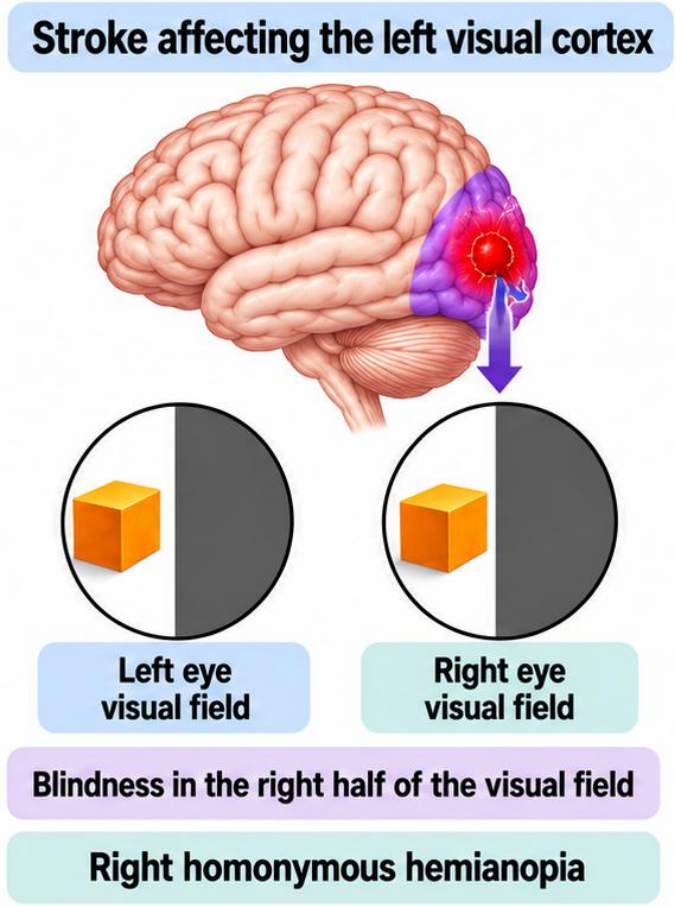


Image Illustrating Loss of Depth Perception. In these examples: **1) Loss of an eye or destruction of one optic nerve** eliminates true depth perception entirely and peripheral vision on damaged side is also affected. **2) Neural destruction** occurring beyond **optic chiasma** → part or all of opposite half of the visual field is lost. **3) Stroke** can affect left visual cortex, which leads to blindness in right half of visual field.



Overview of Hearing

Overview

- **Hearing** depends on the conversion of **mechanical sound waves** → **neural electrical signals** within the **cochlea**, followed by transmission through the **auditory pathway** to the **primary auditory cortex** for **conscious perception** and **interpretation of sound**.

The Ear Has Three Major Areas

- The ear has three major areas:
  - **External (outer) ear:** hearing only
  - **Middle ear (tympanic cavity):** hearing only
  - **Internal (inner) ear:** hearing and equilibrium
- **Receptors** for **hearing and balance** respond to separate stimuli and are activated independently



External and Middle Ear: Sound Conduction

Inner Ear: Cochlear Structure

- **Cavity of cochlea divided into three chambers:**
  - **Scala vestibule:** abuts oval window, contains **perilymph**
  - **Scala media (cochlear duct):** contains **endolymph** – **high K^+**
  - **Scala tympani:** terminates at round window; contains **perilymph** - **normal K^+**
- The **basilar membrane** separates the **cochlear duct** from the **scala tympani**.
- **Reissner membrane** separates the **scala vestibular** from the **scala media**
- The **Organ of Corti** (on the basilar membrane) is the **sensory organ of hearing**.

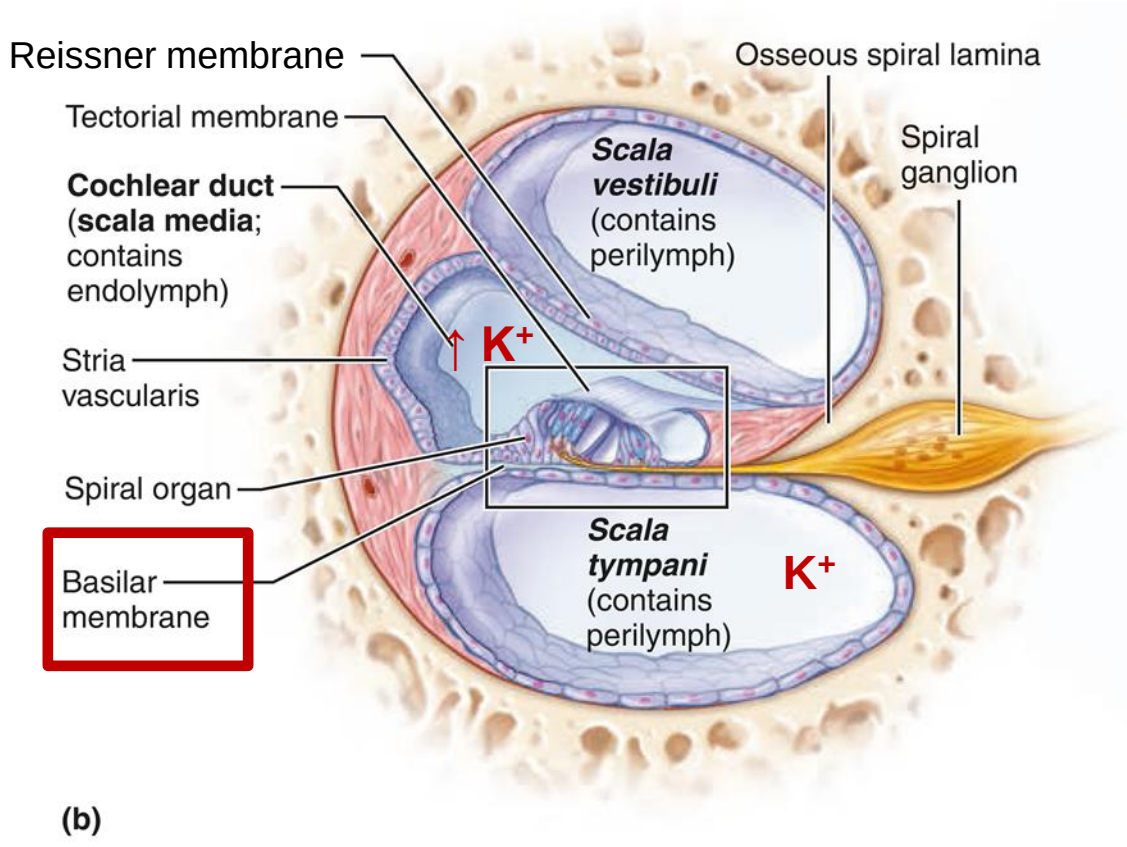


Figure 15.27b Anatomy of the cochlea. *(Adopted from Marieb Anatomy and Physiology, 13th Ed.).*

Inner Ear: Cochlear Structure, Cont.

- **Spiral organ** contains **cochlear hair cells**.
- **Hair cells** are sandwiched between **tectorial and basilar membranes:**
  - Transduce basilar-membrane movement into neural signals
  - Their **stereocilia** project into **endolymph** within the **scala media**
- The **cochlear branch of nerve VIII** runs from spiral organ to the brain

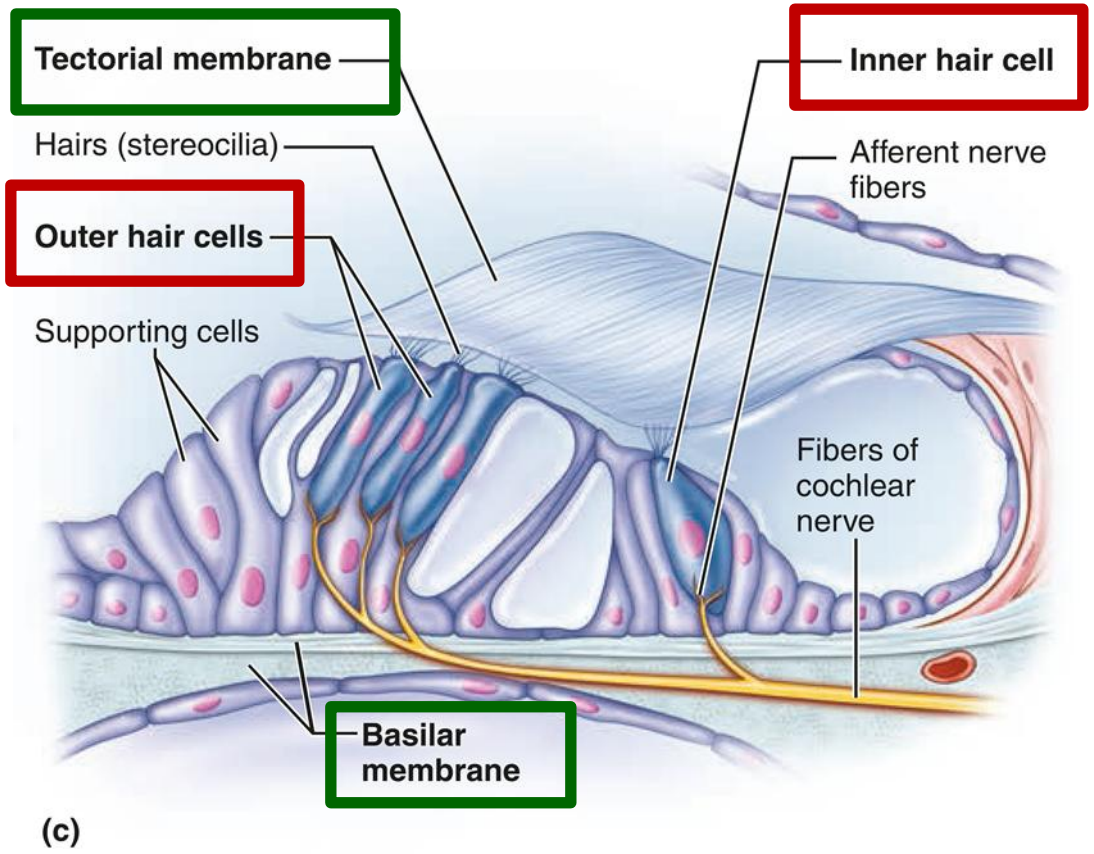


Figure 15.27c Anatomy of the cochlea. *(Adopted from Marieb Anatomy and Physiology, 13th Ed.)*

Inner Ear: Cochlear Structure, Cont.

- **Supporting cells not named:**
  - **Outer pharyngeal**, or **Deiters cells:** support outer hair cells.
  - **Outer pillar cells:** help form the **tunnel of Corti** and support the **organ of Corti**

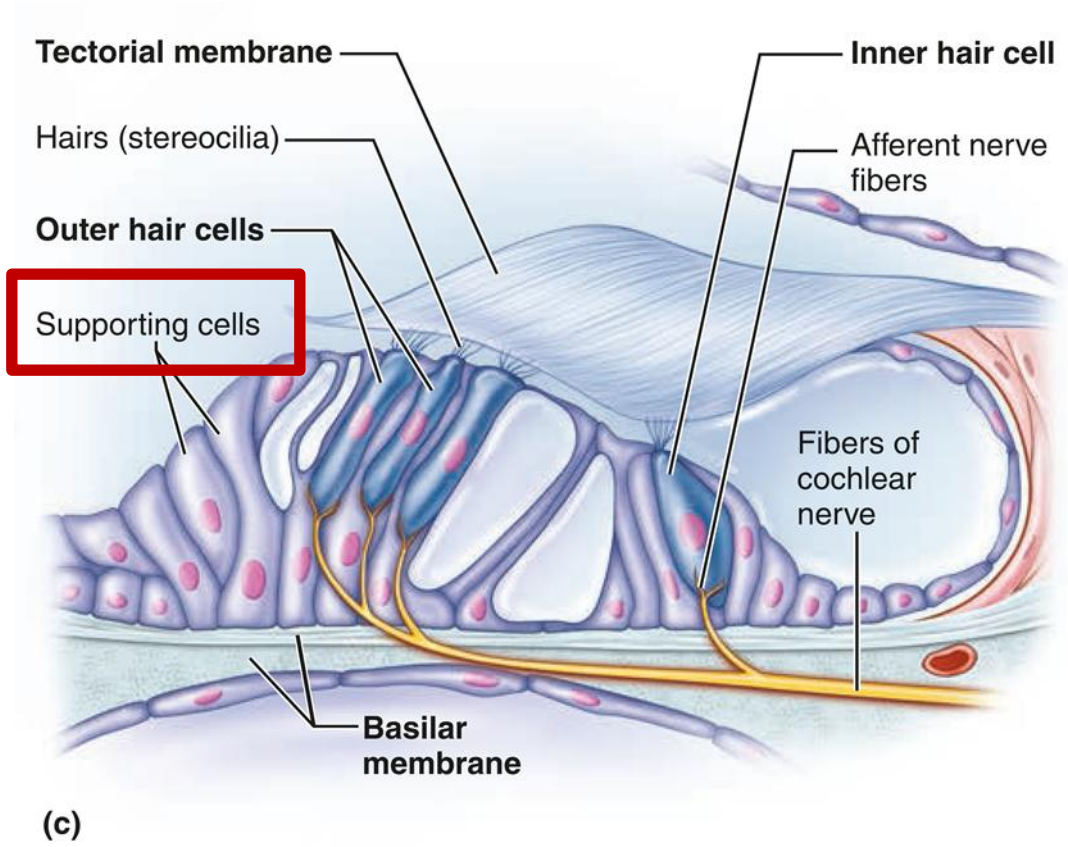


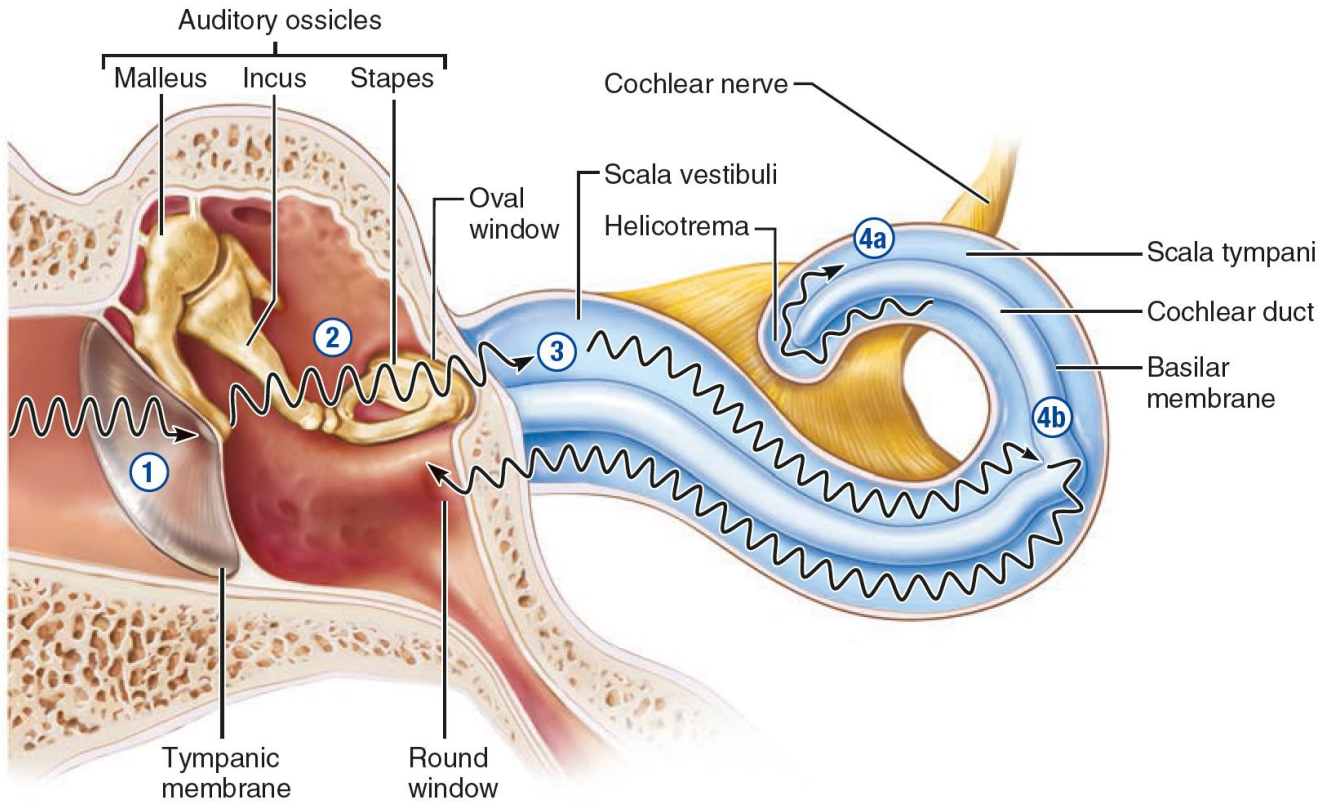
Figure 15.27c Anatomy of the cochlea. *(Adopted from Marieb Anatomy and Physiology, 13th Ed.)*



Physiological Mechanism of Auditory Signal Transduction

Step 1: Mechanical Vibration

- Sound waves → tympanic membrane → ossicles → stapes at oval window → pressure waves in **perilymph** (scala vestibuli).



1. **Sounds waves** vibrate the tympanic membrane.
2. **Auditory ossicles vibrate**. **Pressure** is amplified.
3. **Pressure waves** crated by the stapes pushing on the oval window move through the fluid in the **scala vestibuli**.

4a. Sounds with frequencies **below** the hearing range travel through the helicotrema and **do not** excite hair cells.

4b. Sounds **in the** hearing range go through the cochlear duct, vibrating the basilar membrane and defecting hairs on inner hair cells.

Figure 15.30 Pathway of sound waves. *(Adopted from Marieb Anatomy and Physiology, 13th Ed.)*

Step 2: Basilar Membrane Motion

- **Perilymph** pressure waves deform the **basilar membrane**.
- The **basilar membrane** acts as a **frequency analyzer**:
  - **Base (near oval window): narrow and stiff** → responds best to **high frequencies** (~20,000 Hz)
  - **Apex: wide and flexible** → responds best to **low frequencies** (~20 Hz)
  - **Tonotopic organization** permits us with the ability to **discriminate between sounds with different frequency.**

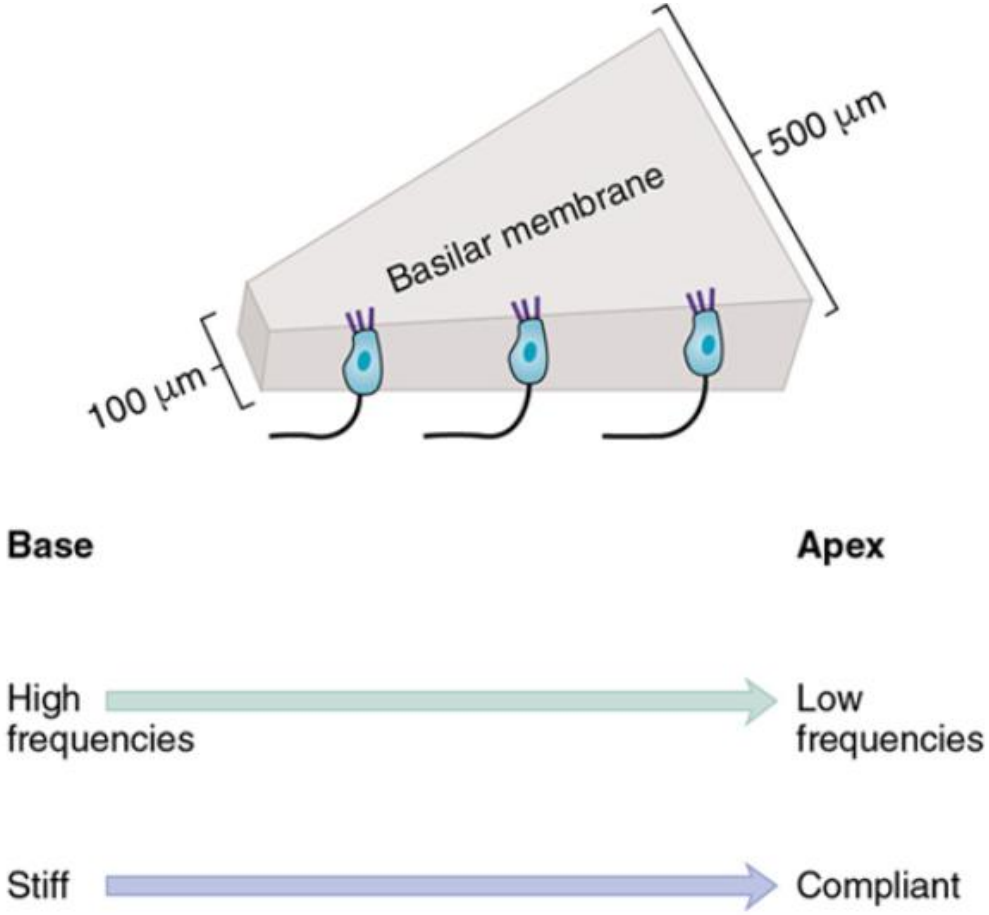


FIG. 3.24 Frequency responses of the basilar membrane.

Step 2: Basilar Membrane Motion, Cont.

- **Hair cells** for **higher frequency** have a smaller interval.
- Hence, **easier to distinguish between various high frequency tones** in comparison to pin-pointing low frequency
- **<u>IMPORTANT:</u>** The **basilar membrane** is in **contact with perilymph** and is **critical to the ability to discriminate between sounds with different frequencies**.

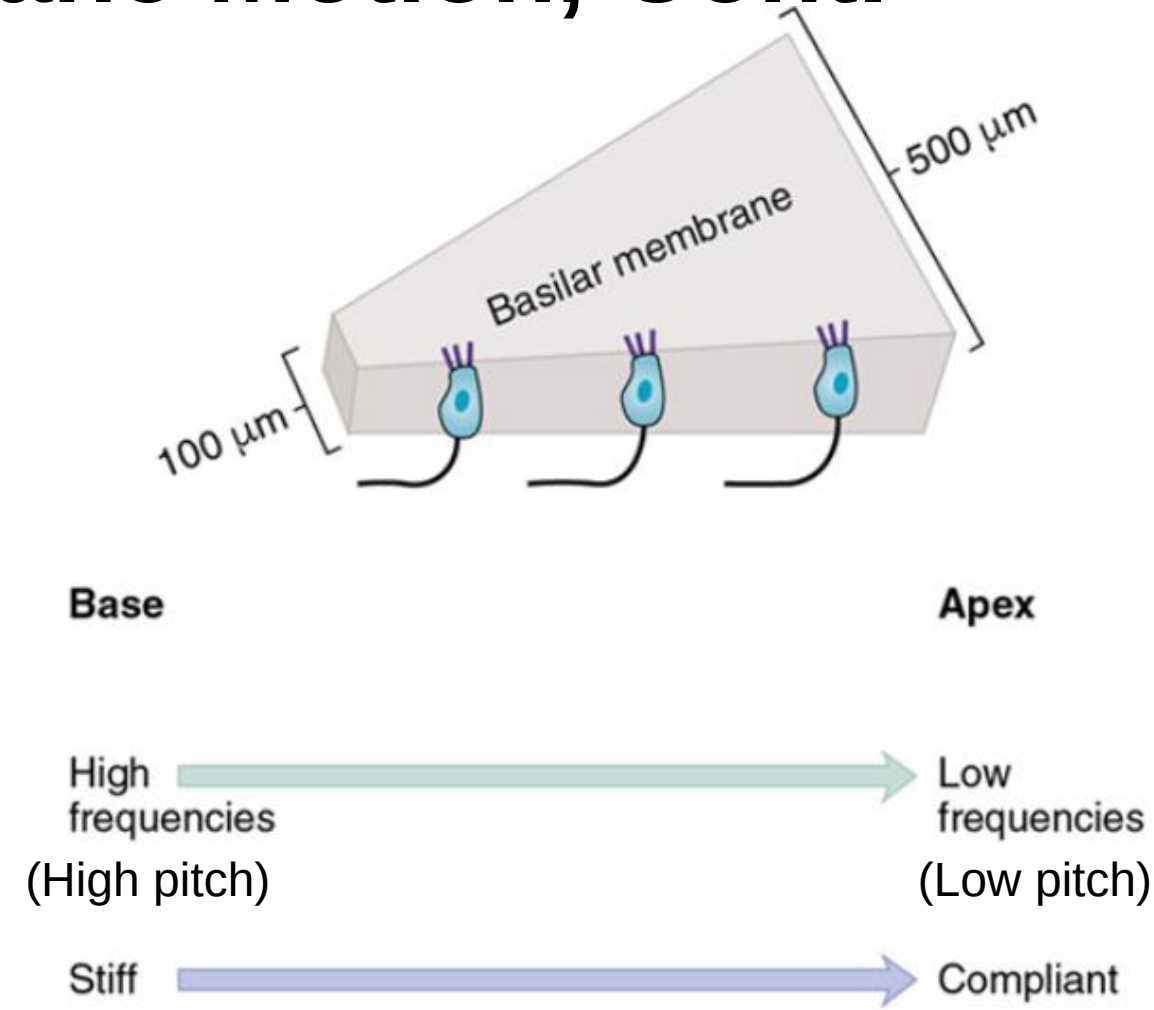


FIG. 3.24 Frequency responses of the basilar membrane.

- **Low-frequency sounds**, such as 20-100 Hz, **CANNOT** move the **short, stiff fibers at the base**
  - These sounds continue to the longer, **floppier fibers at the apex**, where they **vibrate the basilar membrane**
- **Medium-frequency sounds**, such as 800-1000 Hz, **vibrate the basilar membrane near its middle (center)**.
- **High-frequency sounds**, such as 10,000 Hz, **vibrate the basilar membrane near the short stiff fibers at its base**.

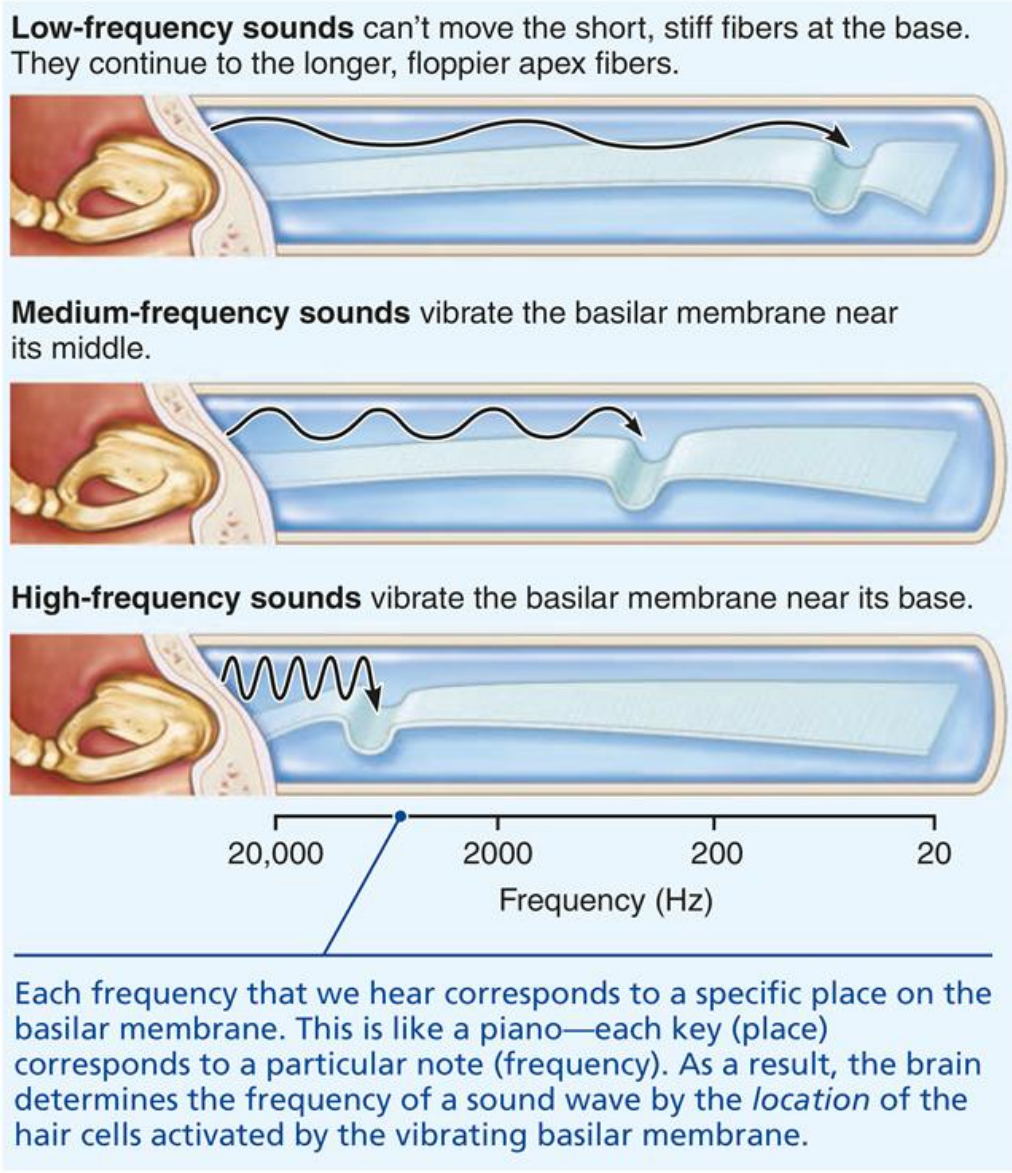


Figure 15.31-3 Basilar membrane function. *(Adopted from Marieb Anatomy and Physiology, 13th Ed.).*

Step 3: Hair Cell Activation

- **The Organ of Corti contains:**
  - **Inner hair cells**: true sensory receptors (afferent)
  - **Outer hair cells**: amplify motion of basilar membrane (efferent modulation)
- The **tectorial membrane** overlies the hair cells.
- **Basilar membrane** movement causes **shearing between the basilar and tectorial membranes**, bending the **stereocilia** on hair cells.

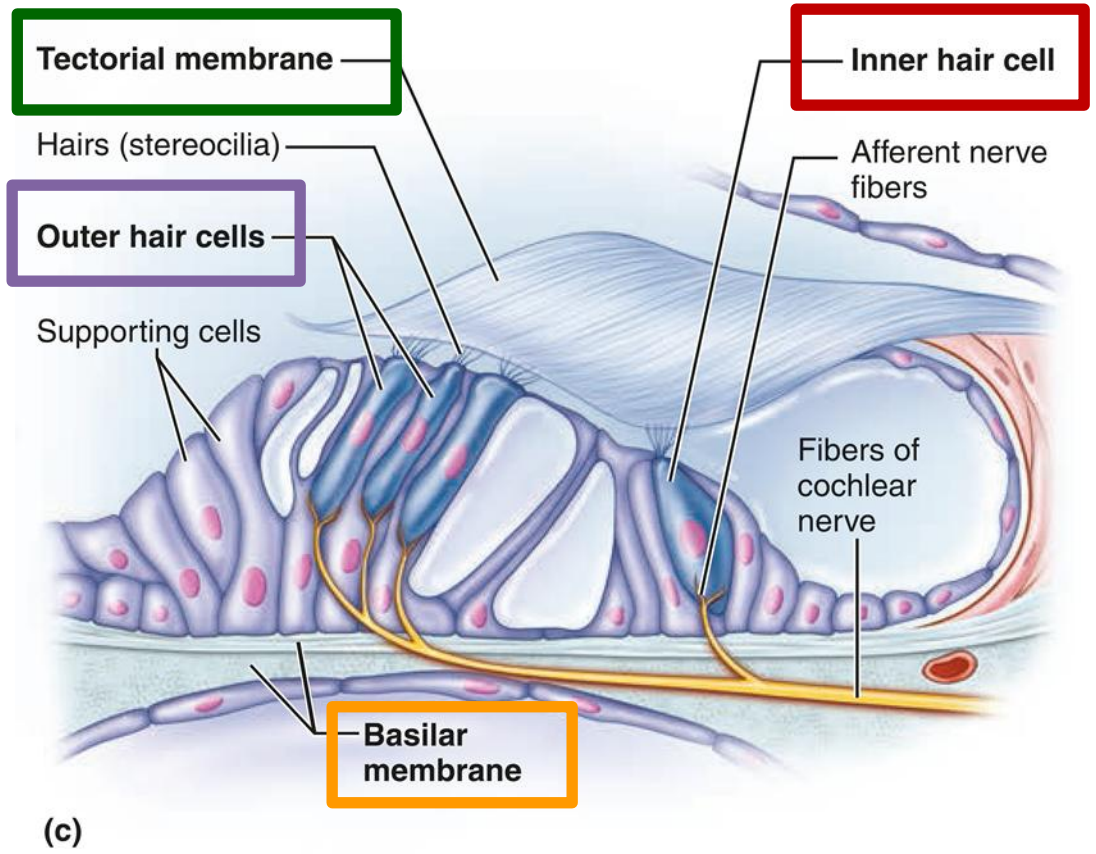


Figure 15.27c Anatomy of the cochlea. *(Adopted from Marieb Anatomy and Physiology, 13th Ed.)*

Step 4: Mechanotransduction

- The **tallest stereocilium** touches the **tectorial membrane**.
- **At rest**, a **few ion channels are open** and the **hair cell is slightly depolarized**.

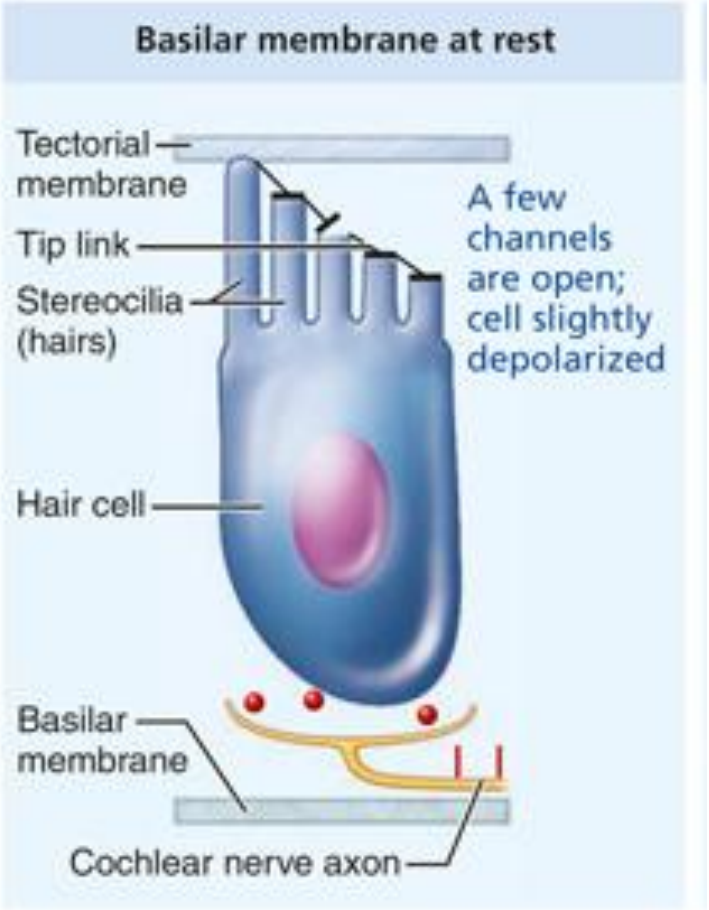


Figure 15.32 Pivoting of stereocilia (hairs) opens or closes mechanically gated ion channels in hair cells. *(Adopted from Marieb Anatomy and Physiology, 13th Ed.)*

Step 4: Mechanotransduction, Cont.

- **Stereocilia deflection toward the tallest cilium opens mechanically gated K^+ channels** at the tips.
- **K^+ influx** (from high - K^+ endolymph) **depolarizes** the hair cell.
- Depolarization opens **voltage-gated Ca^{2+} channels → glutamate release** at the basal synapse onto **afferent fibers** of the **cochlear (spiral ganglion) neurons**.
- As a result, more **action potentials** occur in **cochlear nerve**.

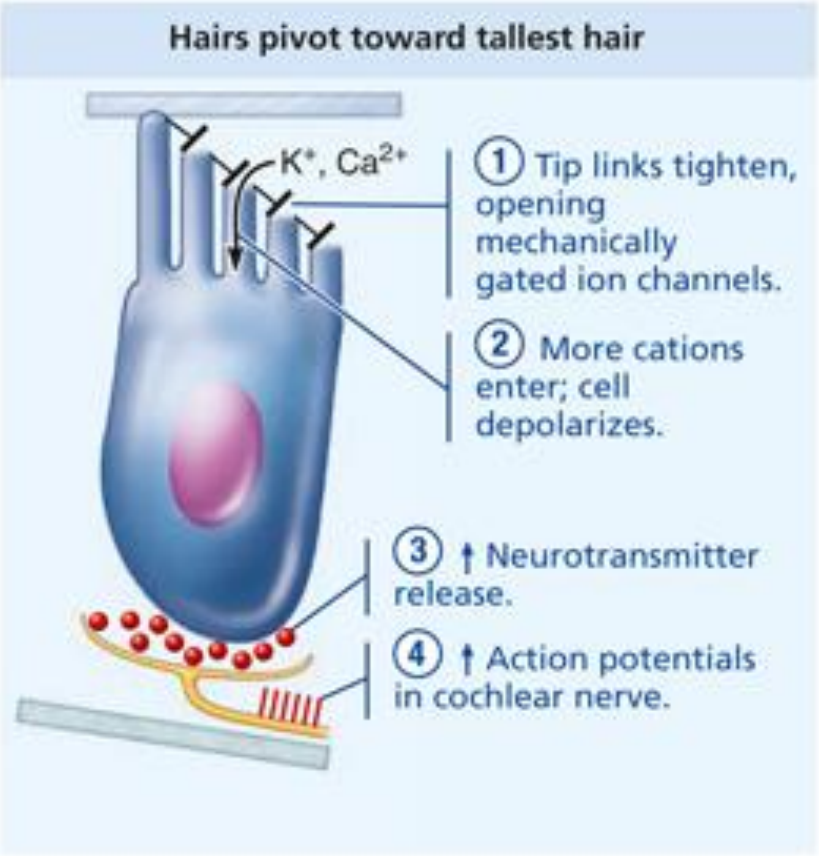


Figure 15.32 Pivoting of stereocilia (hairs) opens or closes mechanically gated ion channels in hair cells. *(Adopted from Marieb Anatomy and Physiology, 13th Ed.)*

Step 4: Mechanotransduction, Cont.

- **Stereocilia deflection away closes** channels → **hyperpolarization** → **decreased neurotransmitter** release in hair cells.

- As a result, **fewer action potentials** occur in the cochlear nerve.



Figure 15.32 Pivoting of stereocilia (hairs) opens or closes mechanically gated ion channels in hair cells. *(Adopted from Marieb Anatomy and Physiology, 13th Ed.)*

Sound Waves

- **Sound waves** travel through the **external auditory canal** and cause the **tympanic membrane** (eardrum) to *vibrate*.
- Movement of the **eardrum** causes *vibration* of the **ossicles** in the **middle ear** (i.e., the malleus, incus, and stapes).
- *Vibration* of the **ossicles** are transferred through the **oval window** and into the **inner ear**.

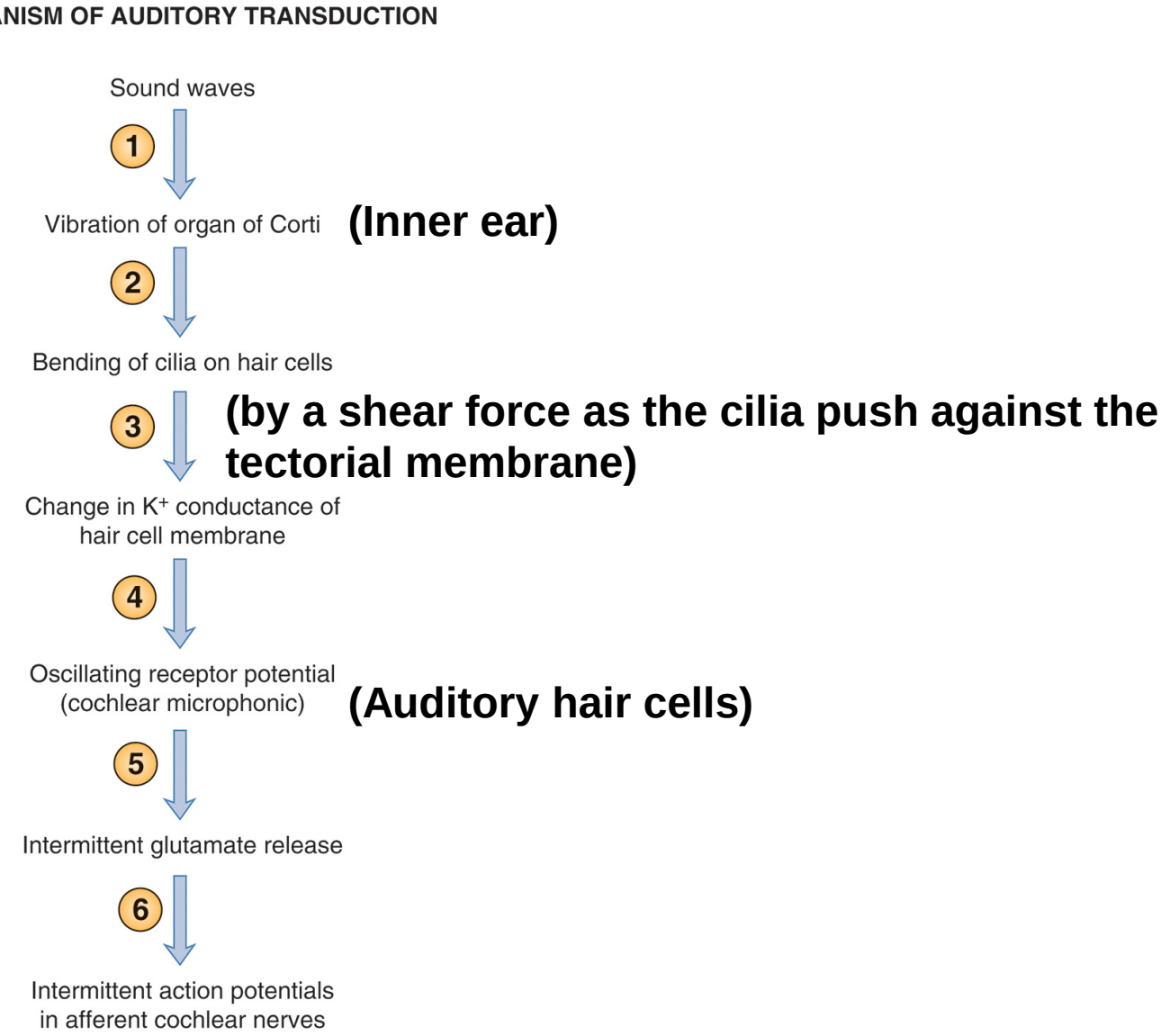


FIG. 3.23 Steps in auditory transduction in hair cells. Circled numbers correspond to steps described in the text.



Auditory Neural Pathway

Auditory Neural Pathway

1. **Cochlear Nerve (CN VIII – Auditory Division)**
   - **Spiral ganglion neurons** form the **cochlear nerve**, carrying frequency-coded information to the brainstem
2. **Cochlear Nuclei (Medulla)**
   - The first central synapse occurs in the **dorsal and ventral cochlear nuclei**
3. **Superior Olivary Complex (Pons)**
   - Receives bilateral input from both ears → critical for **sound localization:**
     - **Medial superior olive:** interaural time differences **(low-frequency sounds)**
     - **Lateral superior olive:** interaural intensity differences **(high-frequency sounds)**

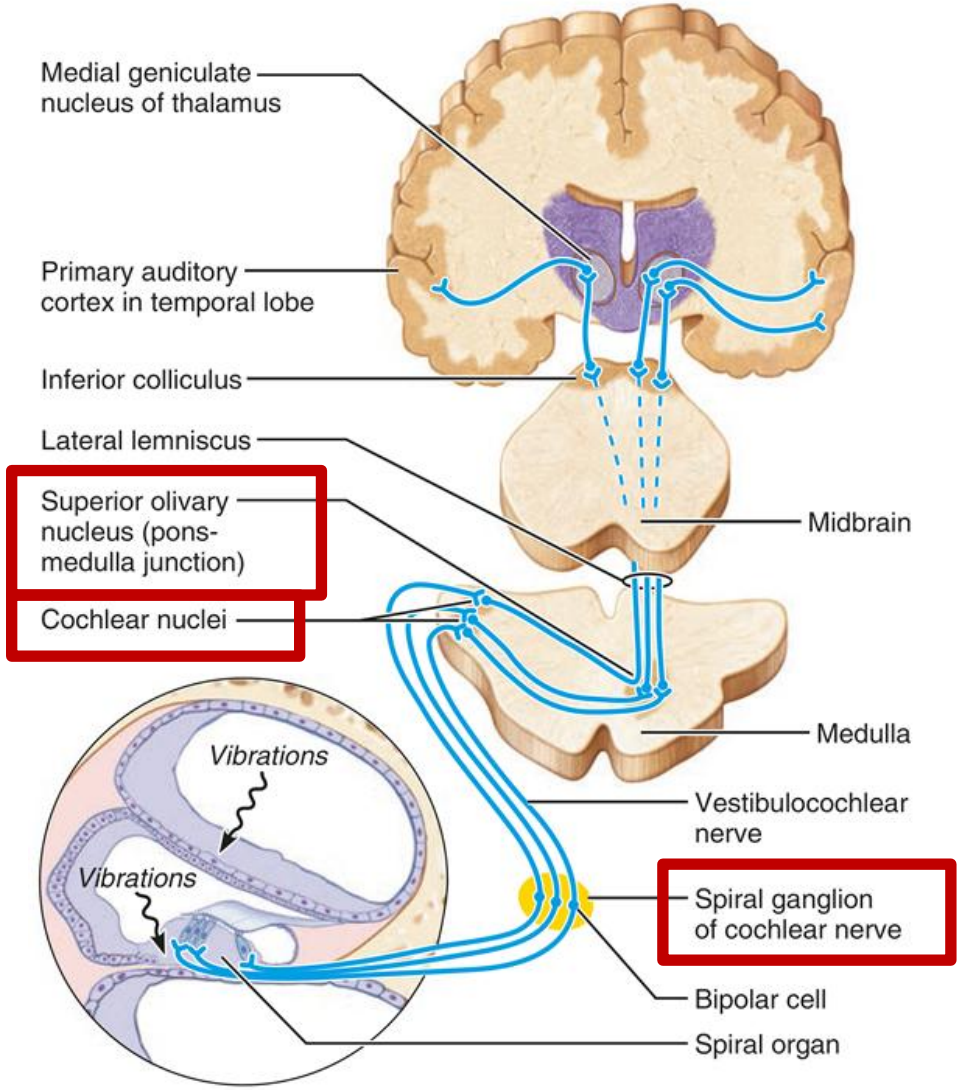


Figure 15.33 The auditory pathway. *(Adopted from Marieb Anatomy and Physiology, 13th Ed.)*

Auditory Neural Pathway, Cont.

4. Lateral Lemniscus (Midbrain)

- Each **lateral lemniscus** carries information derived from both ears; however, input from the contralateral ear predominates
- Carries auditory input from the **cochlear nuclei** and the **superior olivary nuclei** to the **inferior colliculus**

5. Inferior Colliculus (Midbrain)

- Major integration center for auditory reflexes (e.g., orienting to sound)
- Sends auditory information to the **medial geniculate nucleus** (MGN) of the thalamus

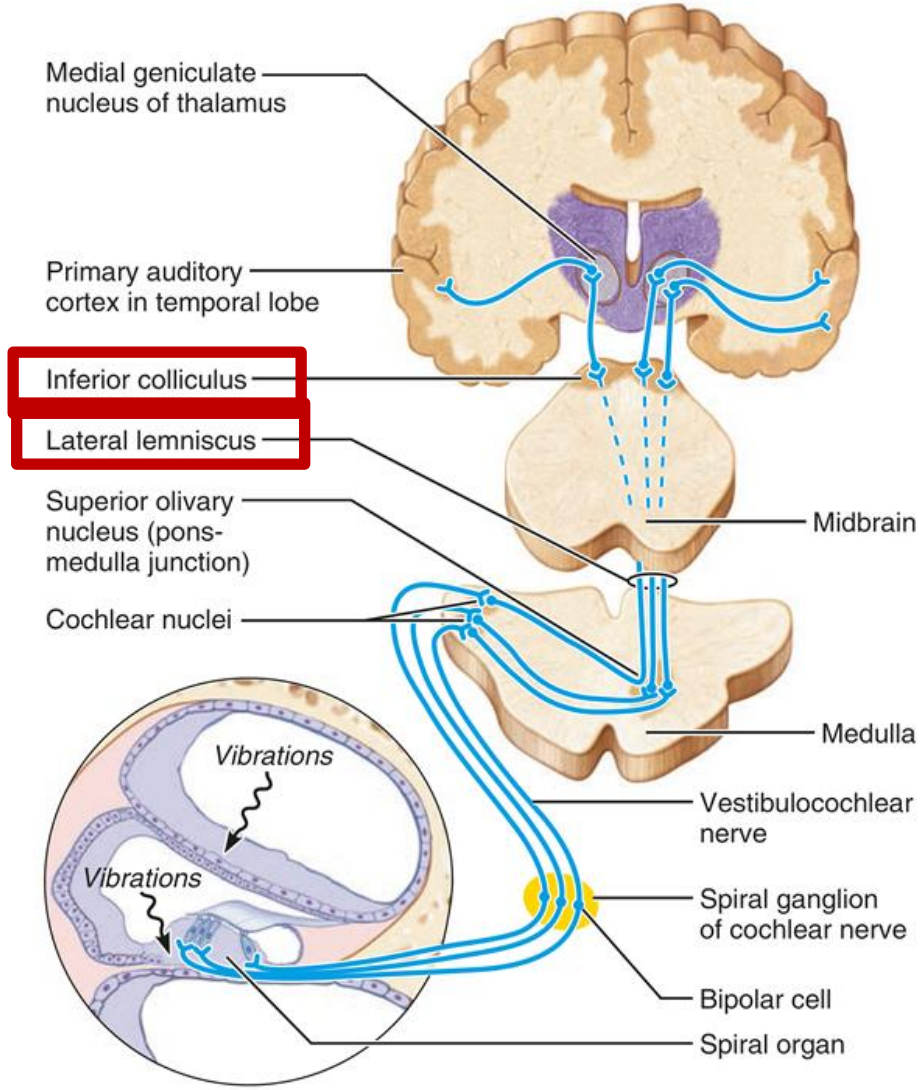


Figure 15.33 The auditory pathway. (Adopted from Marieb Anatomy and Physiology, 13th Ed.

Auditory Neural Pathway, Cont.

6. **Medial Geniculate Body (Thalamus)**
   - Relays auditory information to the **auditory cortex**; functions as a **thalamic "gateway"** for auditory processing
7. **Primary Auditory Cortex (Temporal Lobe)**
   - Located in the **superior temporal gyrus (Brodmann areas 41 and 42, Heschl's gyrus)**
   - **Tonotopic organization is preserved:**
     - **Posterior–medial:** high frequencies
     - **Anterior–lateral:** low frequencies
   - Interprets **sound pitch, intensity, and pattern;** association areas integrate with **language centers (Wernicke's area)**

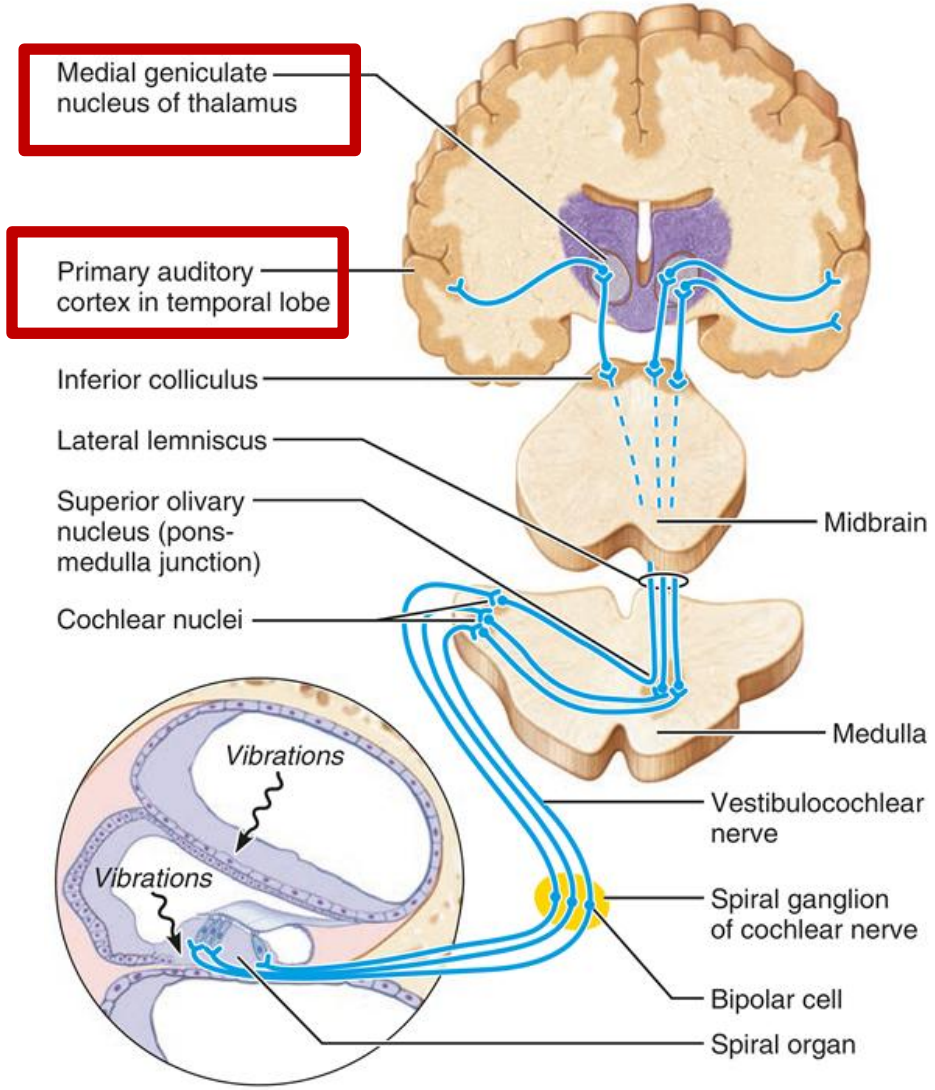
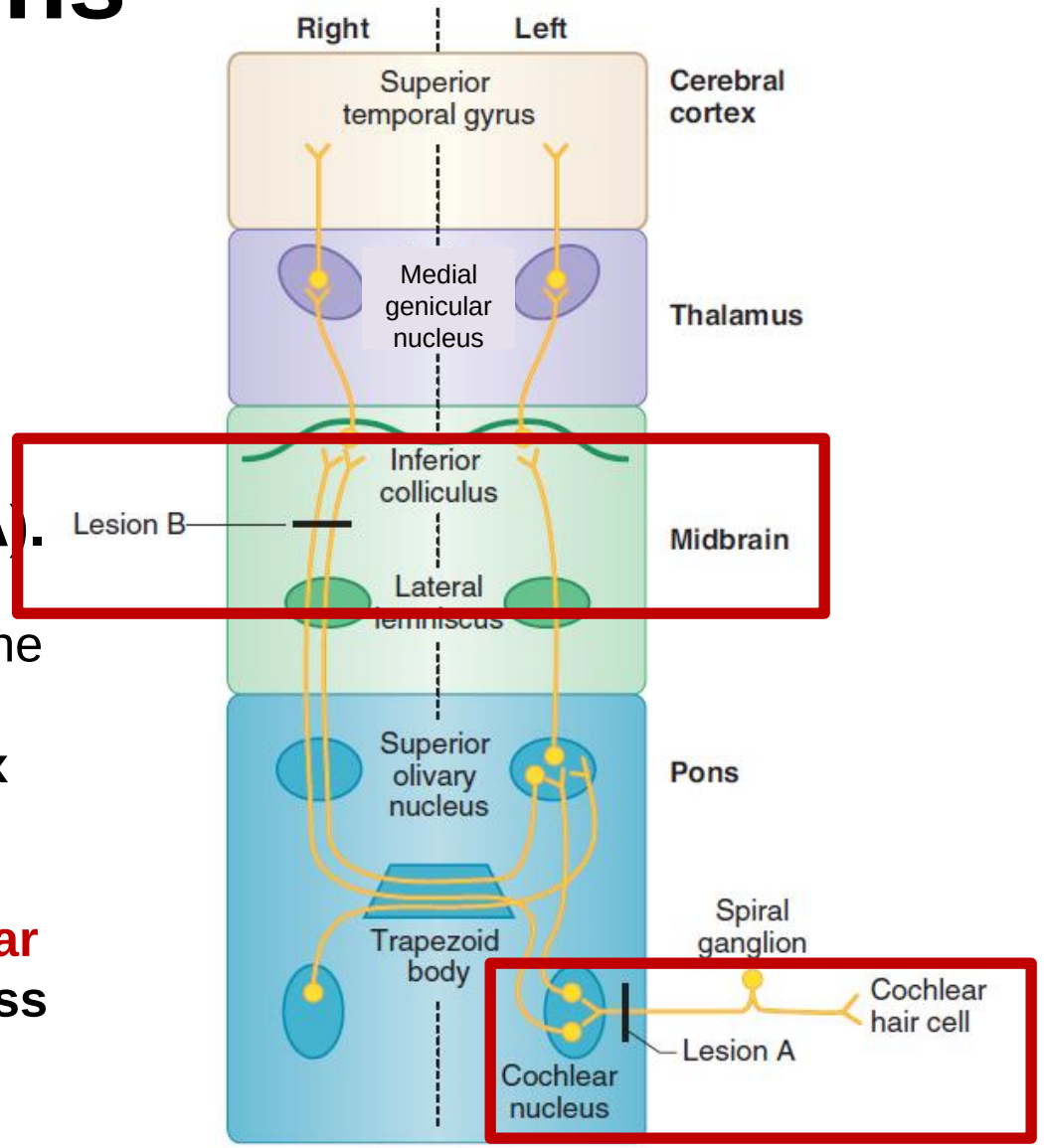


Figure 15.33 The auditory pathway. (Adopted from Marieb Anatomy and Physiology, 13th Ed.).

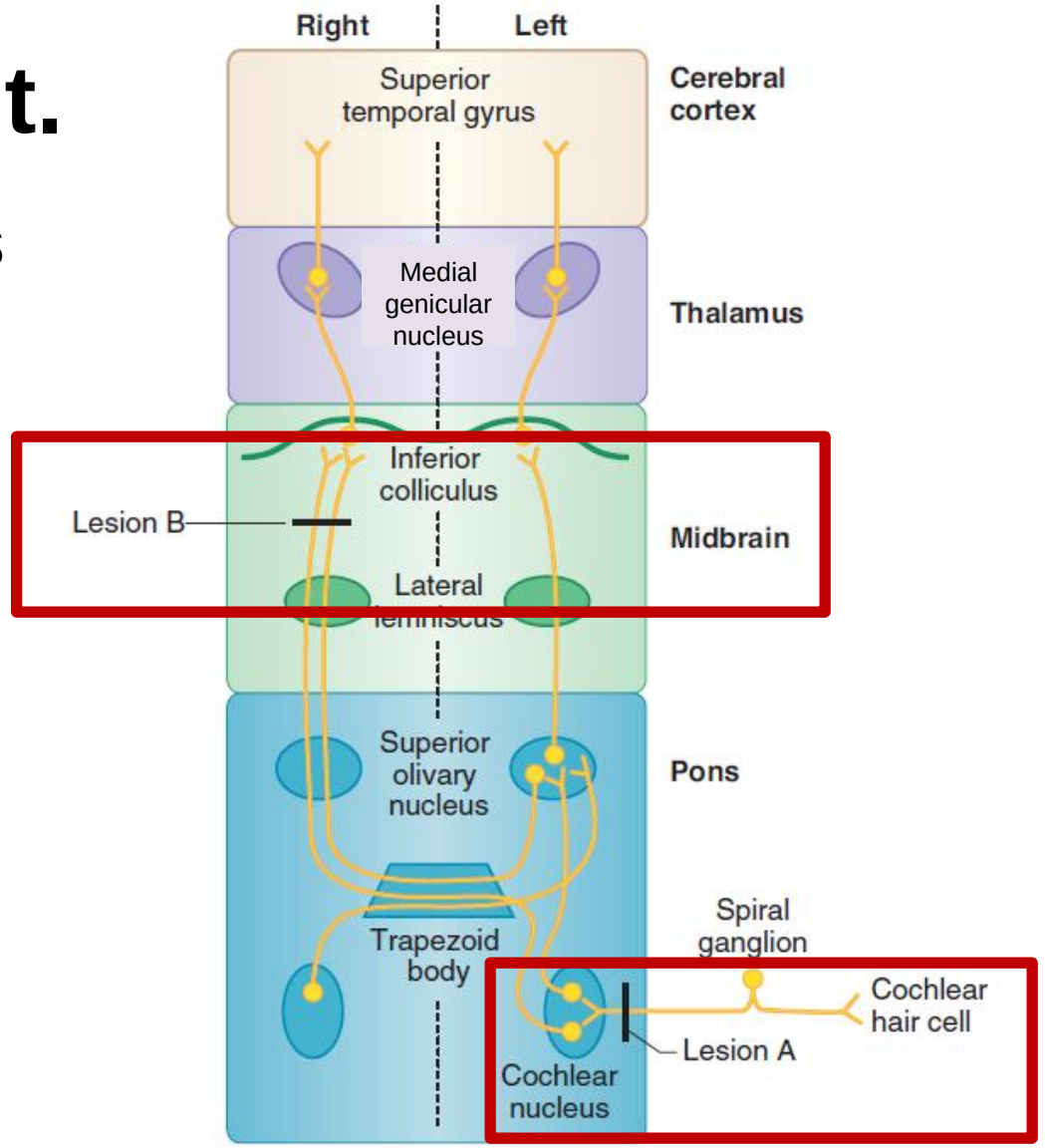
Clinical Correlate: Lesions Causing Hearing Loss

- **Lesions** of the **cochlear part of CN VIII** or **cochlear nuclei** inside the brain stem at the pontomedullary junction result in a profound **unilateral lesion produce ipsilateral deafness**, usually the **sensorineural type (A).**
  - Auditory information from each **cochlea** enters the **ipsilateral cochlear nuclei** before projecting bilaterally through the **superior olivary complex** and **higher auditory pathways**
  - Consequently, a **lesion at or before the cochlear nuclei** can cause marked **ipsilateral hearing loss**
  - Whereas, **unilateral lesions above** this level rarely cause complete unilateral deafness



Clinical Correlate: Lesions Causing Hearing Loss, Cont.

- All other **lesions** to auditory structures in the:
  - **Brain stem** (e.g., inferior colliculus and lateral lemniscus)
  - **Thalamus** (e.g., geniculate body)
  - **Auditory cortex**
- receives bilateral input and therefore result in a **bilateral suppression of hearing** and a **decreased ability to localize a sound source (B)**.



Clinical Correlate: Lesions Causing Hearing Loss, Cont.

- If a patient presents with a **significant hearing loss in one ear**, the **lesion** is most likely in the **middle ear**, **inner ear**, **CN VIII**, or **cochlear nuclei**, and **NOT** at higher levels of the auditory system.
- Again, **lesion of the cochlear of one ear, CN VIII**, or **cochlear nuclei** will cause *unilateral* deafness.

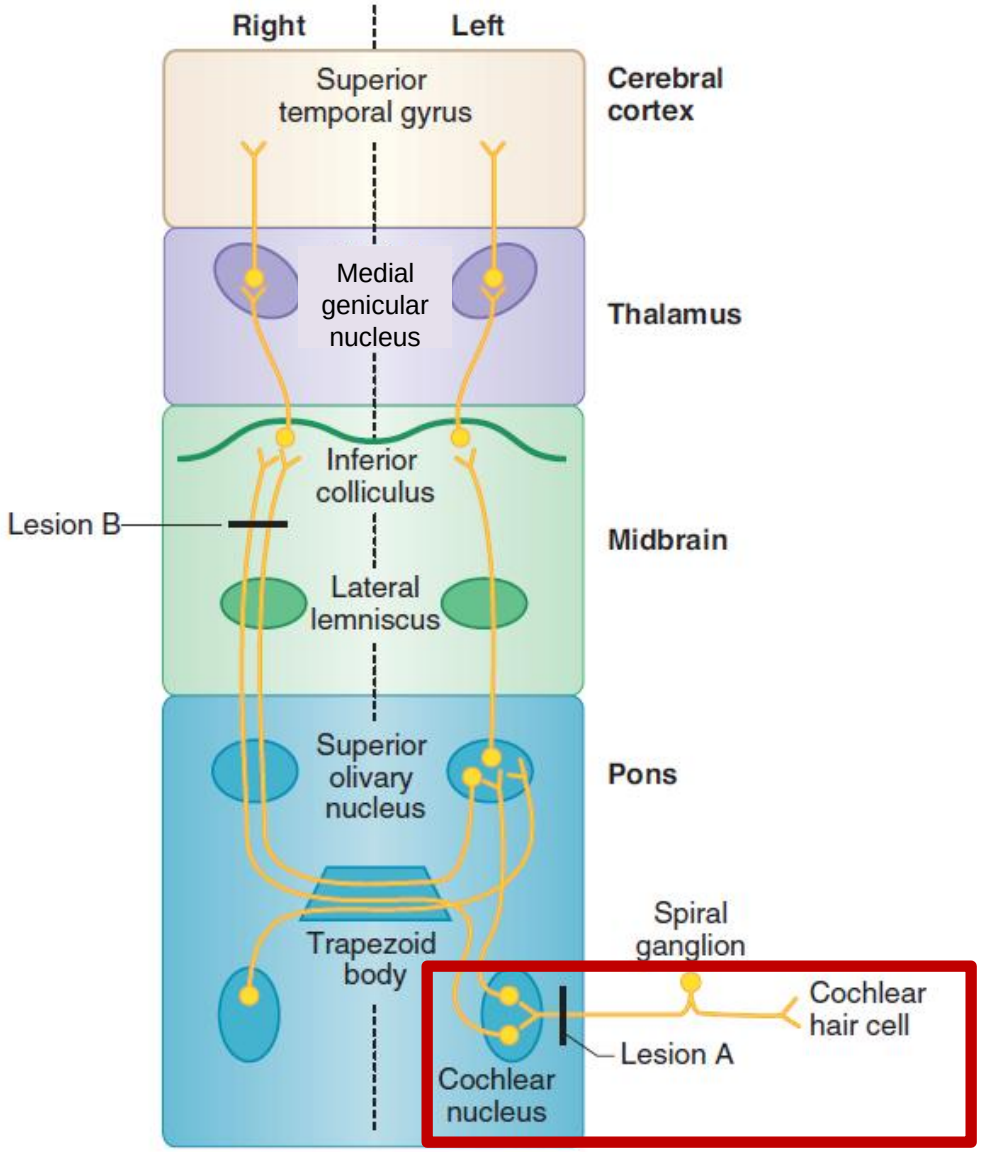


TABLE 2. Clinical Correlations

Lesion Site	Resulting Deficit
Outer/middle ear	Conductive hearing loss
Hair cells or CN VIII	Sensorineural hearing loss
Unilateral cochlear nucleus or auditory nerve	Ipsilateral deafness
Lesion above cochlear nuclei (e.g., cortex)	Bilateral hearing maintained (due to bilateral projections), but impaired sound localization and discrimination

Sound wave → Tympanic membrane → Ossicles → Oval window → Cochlea → Hair cell depolarization → CN VIII → Cochlear nuclei → Superior olive → Lateral lemniscus → Inferior colliculus → Medial geniculate body → **Auditory cortex.**

The End !


HAVE YOUR EYES EVER BEEN CHECKED BEFORE?
NO, THEY'VE ALWAYS BEEN BLUE.

Questions !



Thank You !


HELLO, I'M CALLING TO LET YOU KNOW ABOUT A RECALL OF THE HEARING AIDS YOU RECENTLY PURCHASED...I SAID, I'M CALLING TO... I SAID, I'M CALLING... I SAID...
11-17
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